Drone Pilot and Data Processing

Main Responsibilities of a Drone Pilot

- Operating the drones safely
- Flight palnning
- Pre-flight checks
- Capture aerial images and videos
- Maintain drone equipment
- Troubleshooting problems

Site Preparation:

Define the survey area by insert the coordinates before or outline the area you need on a map or on-site

Pre-flight planning by choose the suitable altitude and the flight path and image overlap (front and side)

Adding GCPs before drone take-off for geo-referencing accuracy of my survey data, GCPs must be clearly marked points placed within the survey area or captured area

Their precise coordinates are surveyed using high accuracy ground surveying equipment like RTK or PPK GPS

Check if any obstacles such as trees, power lines, building or anything can affect safety of drone

Choose a wide-open area for drone takeoff and landing, ensuring the area is clear of obstacles.

Check the weather before take-off, as bad weather conditions like raining or strong winds are very dangerous for the drone flight, fog condition can affect data quality

Flight and Image Capture:

Pre-flight checks before each flight perform a thorough check of the drone

Ensure batteries are fully charged, propollers an intact, the camera is functioning correctly and all sensors are operational

Adjust camera settings like (ISO, shutter speed, aperture and focus) to achieve clear and bright quality images

Fly according to the plan: start the drone and follow the flight path, continuously monitor the drone during flight ensuring it follows the planned path and altitude

Monitor numbers of satellite that the drone receiving their signals

Check the quality of the captured images and make sure they are clear and not blurry

Safe landing after completing the flight mission and capture all images

Data Download to Computer for processing:

After landing, ensure the drone is completely powered off

Remove the SD card from the drone or camera

Use an SD card reader to connect the SD card to your computer

Copy all the data from the SD card to a folder on your computer. Make sure to copy all images and any log files that might be important for processing

After copying, verify that all files have been copied successfully and are not corrupted

After downloading the data, you are ready to begin data processing using specialized Photogrammetry Software

Main Responsibilities of Data Processing

- Data Import and Organization
- Image Alignment and Orientation
- Data Generation
- Quality Control and Accuracy Assessment
- Data Export and Deliverable Creation

Data Import and Organization:

Importing Raw Data, Loading the collected drone imagery (photos or videos), GPS/GNSS data from the drone (if available), and Ground Control Point (GCP) measurements into the processing software.

Data Organization: Structuring the data within the software for efficient processing. This might involve organizing images by flight lines, projects, or dates.

Initial Data Quality Check: Performing a preliminary assessment of the raw images to identify any issues like blurry photos, gaps in coverage, or poor lighting that might affect processing.

Image Alignment and Orientation:

Identifying and marking the GCP locations in multiple images.

Using the measured GCP coordinates to accurately georeference the images

Data Generation:

Depth map estimation, the software estimates depth information for each pixel in the images based on the aligned images and camera parameters

Dense point cloud creation, combining the depth maps from multiple images to generate a dense 3D point cloud

Mesh creation, converting the dense point cloud into a continuous 3D surface mesh

Texture mapping, this makes the model visually realistic and easier to interpret by projecting the original images onto the 3D mesh

Othomosaic, removing geometric distortions from the individual images caused by camera perspective, lens distortion, and terrain relief. This process creates geometrically accurate, map-like images

DEM, extracting elevation information from the dense point cloud or mesh to create a DEM (Digital Elevation Model) representing the surface elevations

DTM, further processing the DEM to remove vegetation and buildings, creating a DTM (Digital Terrain Model) that represents the bare earth surface

Quality Control and Accuracy Assessment:

Visual inspection, reviewing the generated outputs (orthomosaic, point cloud, DEM) visually for any artifacts, errors, or inconsistencies

Accuracy assessment (using Check Points), comparing the derived coordinates of check points (points not used in georeferencing) with their independently surveyed coordinates to assess the accuracy of the processed data

Error reporting and refinement, identifying and documenting any errors or inaccuracies and potentially re-processing data or adjusting processing parameters to improve results

Data Export and Deliverable Creation:

Exporting the processed data in various industry-standard formats (e.g., GeoTIFF for orthomosaics, LAS/LAZ for point clouds, DXF/SHP for vector data, OBJ/PLY for meshes)

Preparing final deliverables for clients, which might include:

- Orthomosaics
- Point Clouds
- DEMs/DTMs
- Contour Lines
- 3D Models
- Reports on accuracy, methodology, and findings.